

Identify the impact of measurement errors on the accuracy of results in practical contexts

Learning intention

- identify the impact of measurement errors on the accuracy of results in practical contexts.

Teach and model

- determining the impact that compounding errors have on financial calculations.
- analysing instruments and methods for measuring in investigations and modelling activities.
- describing settings where measurement errors may impact research results and how measurement data impacted by error can result in biased findings.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.